# Beyond Black Scholes Local Volatility and Stochastic Volatility

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# Introduction

**Option Pricing** Option contracts give the holder the right but not the obligation to buy (or sell) an underlying asset at some point in the future at a predetermined price.

$$C(K,T) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[\max(S_T - K, 0)\big] \qquad C(K,T) = e^{-rT}\int_K^{\infty} (S_T - K)\, f_{\mathbb{Q}}(S_T)\, dS_T$$

**Fourier Transforms** The Fourier transform maps a function from the time domain to the frequency domain

$$\mathcal{F}(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-i\omega t}\, dt.$$

**Characteristic Functions** The characteristic function of a random variable X is

$$\phi_X(t) = \mathbb{E}\big[e^{itX}\big] \qquad \phi_X(t) = \int_{-\infty}^{\infty} e^{itx}\, f_X(x)\, dx$$

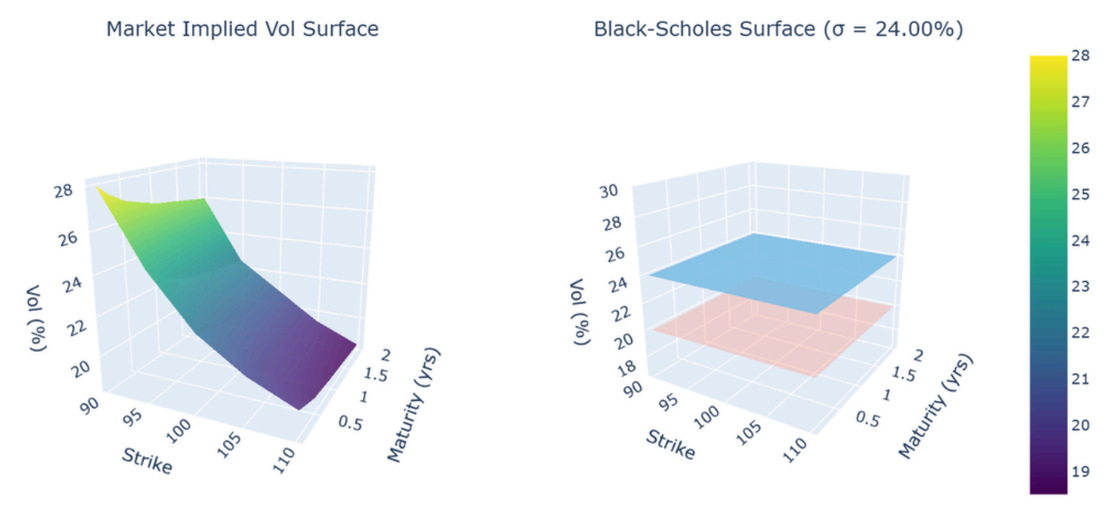
**Stochastic Differential Equations** Most stochastic processes are described by stochastic differential equations (SDEs), typically of the form

$$dX_t = a(t, X_t)\, dt + b(t, X_t)\, dW_t$$

# BSM Model

**The foundational Black–Scholes–Merton model posits that the asset price St follows a geometric Brownian motion under the risk-neutral measure Q:**

$$dS_t = (r - q)S_t\,dt + \sigma S_t\,dW_t,$$




**Black–Scholes constant-volatility surface vs. observed market implied-volatility surface**

The Black–Scholes model assumes a single constant volatility parameter σ for pricing, so the model-implied volatility surface is flat in strike and maturity when expressed as implied vol:

$$\mathbf{BS : (\sigma \mid S, K, r, T) \rightarrow C}$$

**Consequences of the mismatch.**

- **Static mispricing**: A constant-volatility model will underprice some strikes and overprice others.
- **Calibration failure**: You cannot calibrate a single σ to match an entire market surface — at best you can match one point (a particular K, T ).
- **Hedging risk**: Delta/vega hedging strategies derived from Black–Scholes will perform poorly when the true underlying volatility is strike-and time-dependent.

Solving the above SDE we get final BSM equation

$$C = SN(d_1) - Ke^{-rt}N(d_2)$$

**where:**

$$d_1 = \frac{ln\frac{S}{K} + (r + \frac{\sigma_s^2}{2})t}{\sigma_s\sqrt{t}}$$

and

$$d_2 = d_1 - \sigma_s\sqrt{t}$$

**and where:**

$C$ = Call option price

$S$ = Current stock (or other underlying) price

$K$ = Strike price

$r$ = Risk-free interest rate

$t$ = Time to maturity

$N$ = A normal distribution

# Local Volatility Model

The local volatility framework is arguably the simplest extension of Black–Scholes. It assumes that the stock's risk-neutral dynamics satisfy

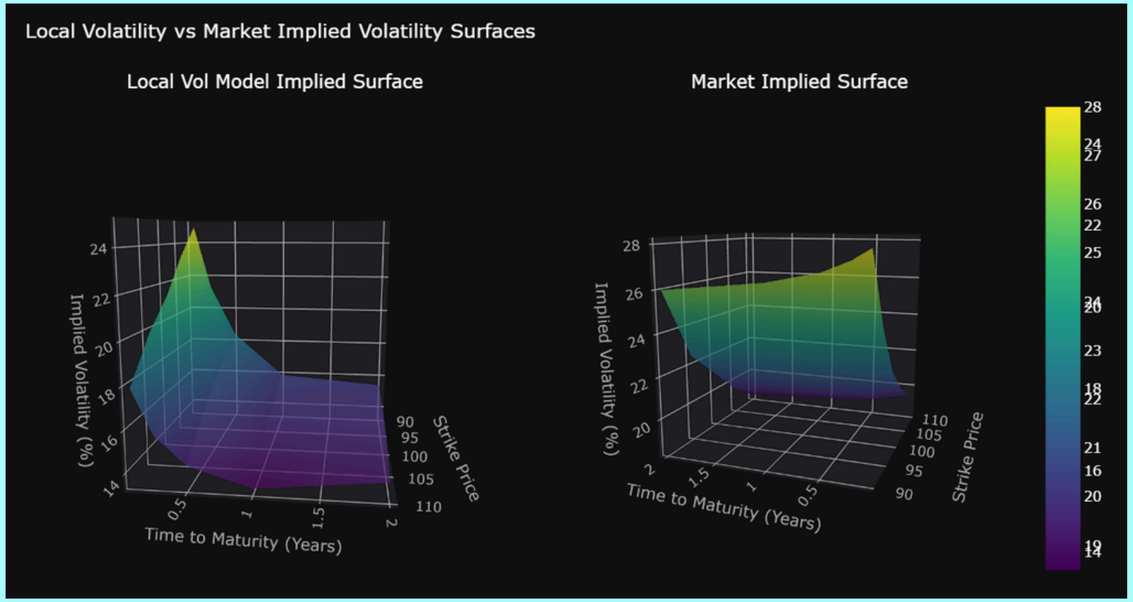
$$dS_t = (r - q)S_t\,dt + \sigma_\ell(t, S_t)\,S_t\,dW_t,$$

## The Dupire Formula

Let C = C(K, T ) be the price of a European call option as a function of strike K and time-to-maturity T . Then the local volatility function satisfies

$$\sigma_\ell^2(T,K) \;=\; \frac{\dfrac{\partial C}{\partial T} + (r-q)\,K\,\dfrac{\partial C}{\partial K} + q\,C}{\dfrac{K^2}{2}\,\dfrac{\partial^2 C}{\partial K^2}}.$$

Given the implied volatility surface we can easily compute the corresponding call option price surface which is the graph of C(K, T) as a function of K and T. It is then clear from The Dupire Formula that we need to take first and second derivatives of this latter surface with respect to strike and first derivatives with respect to time-to-maturity in order to compute the local volatilities.




We observe that the constructed local volatility surface exhibits variation in implied volatility across both strike and time to maturity. However, the parameters of the current model do not yet fit the market-implied volatility surface.

This behaviour is entirely expected: the model has not been calibrated to market data. Until a calibration step is performed, the local volatility surface merely reflects the chosen initial parameterisation rather than the structure implied by observed option prices.

# LV Model Calibration

Given a parameter set θ ∈ Θ ⊂ Rn and a model framework $\mathcal{M}$ we can produce model prices for the given contracts via the pricing map

$$P : \mathcal{M}(\theta, \zeta) \longmapsto \mathbb{R}^m, \qquad P\big(\mathcal{M}(\theta, \zeta)\big) \longmapsto \mathbb{R}^m$$

For a Local Volatility model we may write the risk–neutral dynamics as

$$dS_t = (r - q)S_t\, dt + \sigma_{\text{loc}}(t, S_t)\, S_t\, dW_t,$$

From Dupire Formula we can obtain Exact (theoretical) inversion Dupire - provides an exact relationship between the market call price surface C(K, T ) and the local variance

$$\sigma^2_{\text{loc}}(T, K) = \frac{\partial_T C(K,T) + (r-q)K\, \partial_K C(K,T) + qC(K,T)}{\frac{1}{2}K^2\, \partial_{KK} C(K,T)}.$$

Thus, if one knows the arbitrage-free call-price surface (smooth enough to differentiate), one can recover a local volatility surface σloc(T, K) that exactly reproduces market European option prices:

$$\mathcal{C}_{\text{Dupire}} : \; C(\cdot,\cdot) \longmapsto \sigma_{\text{loc}}(\cdot,\cdot), \qquad P\big(\mathcal{L}(\sigma_{\text{loc}}, \zeta)\big) = \text{market prices (theoretical equality)}.$$

## Calibration and numerical reality

In real markets the call surface is noisy and only observed at discrete (K, T ) points. Thus calibration is an inverse problem and is usually performed by optimisation or numerical inversion

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \; \mathcal{L}\Big(P\big(\mathcal{L}(\theta, \zeta)\big),\; \text{market}(\zeta)\Big) + \lambda \mathcal{R}(\theta),$$

Once a parametrisation is chosen, pricing under the local-vol model requires numerical methods:

- PDE solvers (finite-difference / Crank–Nicolson) for the forward/backward Kolmogorov PDE
- Monte Carlo simulation (Euler discretisation with local σloc(t, S))
- Transform / eigenfunction methods in special cases

**Practical remarks**

- **Fit**: Local volatility can be calibrated to exactly match European option prices
- **Stability**: Dupire inversion is sensitive to noise; smoothing and regularisation are essential
- **Dynamics**: While vanillas are matched, path-dependent products may be mispriced
- **Implementation**: Results depend strongly on grid design, interpolation, and numerical schemes

$$\text{market prices} \xrightarrow{\text{interpolate / smooth}} C(K,T) \xrightarrow{\text{Dupire / optimization}} \widehat{\sigma}_{\text{loc}}(T,K) \xrightarrow{\text{PDE/MC}} \widetilde{P}\big(\mathcal{L}(\widehat{\sigma}_{\text{loc}}, \zeta)\big) \approx \text{market prices}.$$

# Heston Volatility Model

Under the usual risk-neutral measure the Heston model for the asset price St and instantaneous variance vt is

$$dS_t = (r - q)\, S_t\, dt + \sqrt{v_t}\, S_t\, dW_t^{(s)},$$

$$dv_t = \kappa(\theta - v_t)\, dt + \xi\sqrt{v_t}\, dW_t^{(vol)},$$

with correlation $d\langle W^{(s)}, W^{(vol)}\rangle_t = \rho\, dt$

where W (s)t and W (vol)t are standard Q-Brownian motions with constant correlation coefficient ρ

Whereas the local volatility model is a complete model, Heston's stochastic volatility model is an incomplete model. This should not be too surprising as there are two sources of uncertainty in the Heston model

## The Pricing PDE

Because Heston's model is incomplete it is not possible to price options using the replication arguments that apply to complete market models. Instead, one assumes Q-dynamics and then prices all securities using this EMM. (The model parameters are chosen by calibrating model prices to observable market prices.) The pricing PDE that the price, C(t, St, σt), of any derivative security must satisfy in Heston's model is given by

$$\frac{\partial C}{\partial t} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + \rho\sigma\gamma S \frac{\partial^2 C}{\partial S \partial \sigma} + \tfrac{1}{2}\gamma^2 \frac{\partial^2 C}{\partial \sigma^2} + (r - q) S \frac{\partial C}{\partial S} + \kappa(\theta - \sigma)\frac{\partial C}{\partial \sigma} = rC.$$

## Pricing Function

Given a model class M and a parameter set θ ∈ Θ ⊂ Rd we obtain a pricing map For the Heston model we may write

$$P:\ \mathcal{M}(\theta, \zeta) \longmapsto \mathbb{R}^m.$$

$$\theta = (\kappa,\, \theta,\, \xi,\, \rho,\, v_0), \qquad \mathcal{M} = \mathcal{H}\ (\text{Heston}).$$

When using a numerical solver or simulation we denote the approximate pricing map by $\widehat{P}(\mathcal{H}(\theta, \zeta)) \approx P(\mathcal{H}(\theta, \zeta))$

## Model Calibration

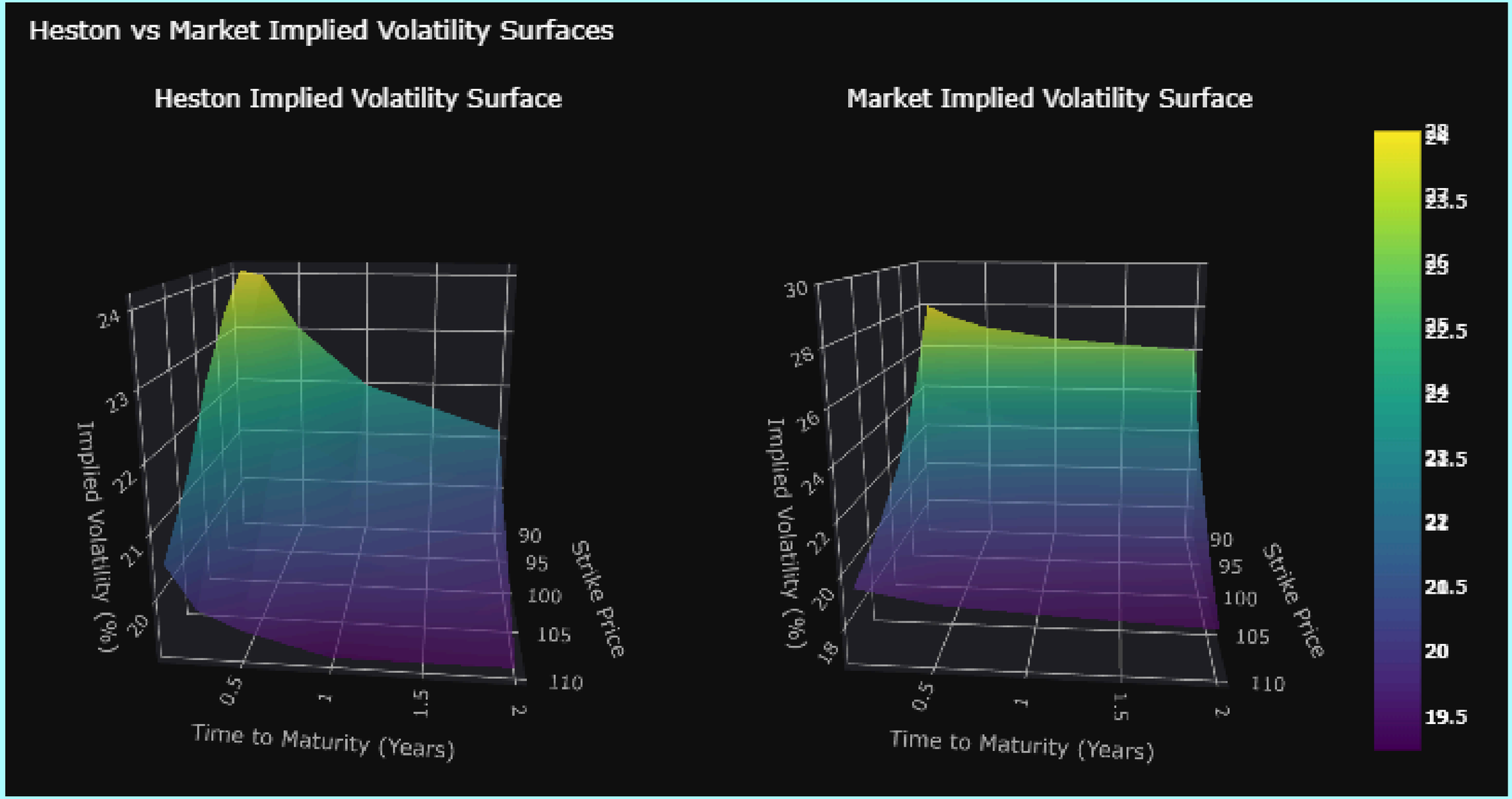
Once a pricing routine (closed–form, quasi–closed, simulation, or PDE) is chosen, calibration finds parameter values θ such that model prices match market prices as closely as possible. A common formulation is the minimization problem

$$\theta^* = \underset{\theta \in \Theta}{\operatorname{argmin}}\ \delta\big(\widehat{P}(\mathcal{H}(\theta, \zeta)),\ P^{\mathrm{MKT}}(\zeta)\big),$$

where δ(·, ·) is a chosen distance (for example weighted squared errors in price or implied volatility space) and P MKT(ζ) are the observed market prices.

Interpretation: θ∗ are the Heston parameters (κ, θ, ξ, ρ, v0, K, T ) to use based on market prices.

# Heston Vol Surface



# Pricing Exotic Options

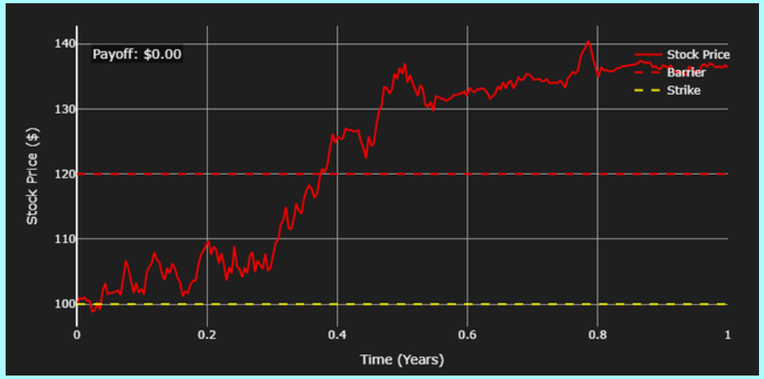
## Barrier Options

A knock-out barrier option requires the price path to not hit a certain price threshold otherwise the payoff is zero.
Let B be a barrier level and τ the first hitting time of St to B

$$\tau = \inf\{t \geq 0 : S_t = B\}.$$

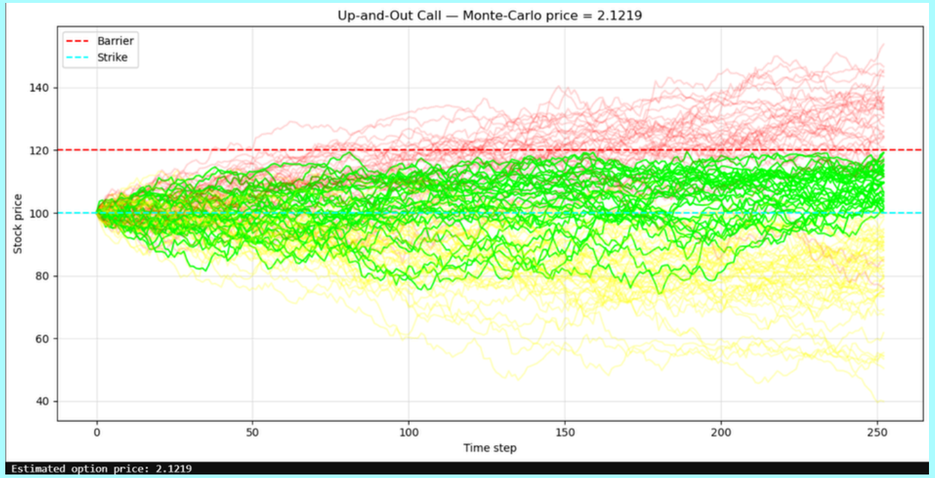
For an up–and–out barrier call option with maturity T the payoff is knocked out once the barrier is hit. Its price under the Heston model can be written as

$$V(S_0, v_0; \theta) = \mathbb{E}^{\mathbb{Q}}\left[e^{-rT}(S_T - K)^+ \mathbf{1}_{\{\tau > T\}} \mid S_0, v_0\right],$$

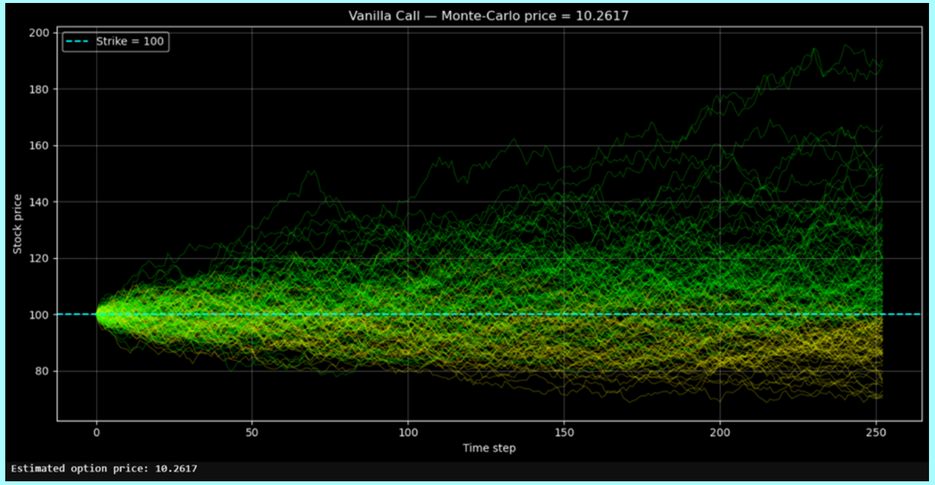



Closed–form solutions for such path–dependent claims are generally unavailable under stochastic volatility. A natural approach is therefore Monte Carlo simulation:

- Simulate joint paths (St, vt) under the risk–neutral measure.
- For each path, monitor whether the barrier B is breached.
- Compute the discounted payoff (St − K)+ if τ > T , and zero otherwise.
- Average over many paths to approximate V (S0, v0; θ).




**Monte Carlo simulation for Barrier Option**




**Monte Carlo simulation for Vanilla Call Option**

# Carr-Madan (Lesss go!)

Monte carlo is a slow calibration techinque, therefore we can use concepts of FFT

But, the problem is we do not know how to solve call price intergral in time domain so what we can do is apply foriuer transform go from time domain to frequency domain and then apply inverse Fourier transform to go back to time domain

$$C(K,T) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\max(S_T - K, 0)]$$

where the expectation is defined as

$$\mathbb{E}^{\mathbb{Q}}[\max(S_T - K, 0)] = \int_{-\infty}^{\infty} \max(S_T - K, 0) f_{S_T}(x) \backslash \mathrm{d}x = \int_{K}^{\infty} (S_T - K) f_{S_T}(x) \backslash \mathrm{d}x.$$

What if we were to apply a Fourier transform to this call option value?
Well we run into an issue – the standard call option pricing function is NOT square-integrable which is necessary for the Fourier transform. This is due to the log-strike approaching a constant at −∞ which is not zero.

## Damping Factor $e^{\alpha k}$

By introducing a damping factor, Carr–Madan enable square-integrability of the pricing function and thus the Fourier transform applies.

Let, damped call price $c_T(k) = e^{\alpha k} C_T(k)$

The Fourier transform of the damped call price is: $\psi_T(v) = \int_{-\infty}^{\infty} e^{ivk}\, c_T(k)\, \mathrm{d}k$

And the inverse transform gives us the original call price:

$$C_T(k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-ivk}\, \psi_T(v)\, \mathrm{d}v$$

### Key Equation

The key equation for the Fourier transform of the damped call price in terms of the characteristic function is:

Starting with the Fourier transform of the damped call price:

$$\psi_T(v) = \int_{-\infty}^{\infty} e^{ivk} c_T(k) dk = \int_{-\infty}^{\infty} e^{ivk} e^{\alpha k} C_T(k) dk$$

Substituting the call price formula:

$$\psi_T(v) = \int_{-\infty}^{\infty} e^{ivk} e^{\alpha k} e^{-rT} \int_{k}^{\infty} (e^y - e^k) f_{S_T}(y) dy\, dk$$

Changing order of integration:

$$\psi_T(v) = e^{-rT} \int_{-\infty}^{\infty} \int_{-\infty}^{y} e^{ivk} e^{\alpha k} (e^y - e^k) dk\, f_{S_T}(y) dy$$

Evaluating the inner integral:

$$\psi_T(v) = e^{-rT} \int_{-\infty}^{\infty} \left[ \frac{e^y e^{(\alpha+iv)k}}{(\alpha+iv)} - \frac{e^{(\alpha+iv+1)k}}{(\alpha+iv+1)} \right]_{-\infty}^{y} f_{S_T}(y) dy$$

Simplifying:

$$\psi_T(v) = e^{-rT} \int_{-\infty}^{\infty} e^y \left( \frac{e^{(\alpha+iv)y}}{(\alpha+iv)} - \frac{e^{(\alpha+iv+1)y}}{(\alpha+iv+1)} \right) f_{S_T}(y) dy$$

Recognizing the characteristic function:

$$\psi_T(v) = \frac{e^{-rT}}{(\alpha+iv)(\alpha+iv+1)} \phi_T(v - (i\alpha+1)) S_0^{iv+\alpha+1}$$

where $\phi_T(u)$ is the characteristic function of the log-price process.

Fortunately, the characteristic equation is known for the Heston model

$$\phi_{X_T}(u) = \exp\left( \mathrm{i}u(\ln S_0 + rT) + \frac{\kappa\theta}{\sigma^2} \left( (\kappa - \rho\sigma \mathrm{i}u - d)T - 2\ln \frac{1 - ge^{-dT}}{1-g} \right) + \frac{v_0}{\sigma^2} \frac{1 - e^{-dT}}{1 - ge^{-dT}} (\kappa - \rho\sigma \mathrm{i}u - d) \right)$$

# Price Adjustment

## Counterparty Risk Adjustment

- Credit Value Adjustment (CVA) accounts for counterparty default risk.
- Debt Value Adjustment (DVA) accounts for own default risk.
- These adjustments reduce the price when default risk is higher.

## Liquidity Adjustment

- Exotic options are less liquid than vanilla options
- Wider bid–ask spreads reflect higher hedging costs
- Size of adjustment depends on: Option complexity, Market conditions, Position size

## Profit Margin

- Trading desks add profit margin to cover:

1. Operating costs
2. Capital charges
3. Return on equity requirements

- Typical margins range from 2–10% depending on: Client relationship, Competition, Market conditions

After calculating the base exotic option price using our calibrated model, several adjustments are typically made

$$Q = P + CVA + DVA + L + M$$

where:

- P is the theoretical price from the calibrated model
- CVA is the credit value adjustment
- DVA is the debt value adjustment
- L is the liquidity adjustment
- M is the profit margin

# Thank you!

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